Pratyusha Adibhatla

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H4 EAD — Open to: C2C / Contract / W2 — Current Location: Long Island,NY — Relocation: Yes

Summary

Data Engineer with a MS in Data Science and expertise building scalable ETL pipelines and cloud-based platforms across Azure and AWS. Proficient in Python and advanced SQL, specializing in transforming complex healthcare and financial datasets into production-ready semantic layers. Skilled at optimizing workflows to ensure data integrity, reduce latency, and power real-time analytics.

Experience

Research Assistant - Data Engineer March – August 2025
UCSD Wireless Communication, Sensing, and Networking Lab *San Diego, CA*

- Deployed an Azure-native ingestion framework using Python and SQLite to manage HDF5, CSV, DAT, and PCD datasets stored in Azure Blob Storage, reducing discovery time to under 3 minutes through indexed metadata queries.
- Initiated an automated batch ETL workflows using Azure Pipelines to orchestrate metadata extraction, schema validation, and data quality checks for 5GB+ datasets per batch.
- Implemented data lineage and feature retrieval systems within the Azure ecosystem to enable version-controlled querying and downloading of datasets with comprehensive change tracking.

Research Assistant - Data Engineer – Retrieval Systems Sep 2024 – Feb 2025
University at Buffalo *Buffalo, NY*

- Established a vector search data pipeline to ingest and structure unstructured data (video, image, text) by converting video to transcripts, images to OCR text, and documents to cleaned text for low-latency retrieval utilizing RAG.
- Optimized data indexing workflows in ChromaDB by designing metadata schema, regenerating embeddings, and refreshing vector indexes when new data was added to ensure consistency across modalities.

Student Assistant – Data Engineering & Analytics Sep 2023 – Jan 2024
University of Texas at Arlington *Arlington, TX*

- Designed a structured SQL database(Star Schema) to replace manual tracking sheets, implementing strict referential integrity constraints across 10+ tables to eliminate data redundancy and ensure consistent inventory reporting.
- Built Python-based batch ETL jobs to ingest and sanitize raw inventory data, implementing custom logic to handle schema variations and null values from legacy Excel inputs, automating what was previously a manual data entry workload.
- Spearheaded a complex SQL scripts using window functions to aggregate historical usage data across 200+ facility items, optimizing query performance to generate accurate demand forecasting reports for inventory procurement planning.

Associate Data Engineer May 2021 - Jun 2022
Cognizant (Client: Liberty Mutual Insurance) *Coimbatore, India*

- Enhanced & supported production batch ETL pipelines on **AWS EC2** processing 1M+ insurance claims and payment transactions monthly; ensured SLA adherence by performing root cause analysis on recurring job failures(15–20 weekly), analyzing S3 dependencies, database locks, and logs to standardize recovery steps.
- Optimized SQL transformation logic across **Amazon RDS** transactional tables and **Redshift** analytical workloads to reduce reconciliation runtime, while tracing end-to-end transaction flows to resolve cross-system inconsistencies and stranded payment records involving **DynamoDB** metadata tables.
- Automated data validation checks reconciling queries to identify schema drift and null anomalies, refining checkpoints, strengthening source-to-target reconciliation for 200–300 financial transactions per batch to ensure payment accuracy and audit compliance.
- Executed controlled Transaction Data Update (TDU) jobs and production data corrections under governance controls, managing data-related incidents through **Jira** workflows and collaborating cross-functionally during production releases.
- Created structured runbooks embedding **SQL** validation queries for TDU handling and failed job triage, documenting expected row counts, financial control totals, validation thresholds (null %, variance limits), common failure scenarios, and standardized rerun procedures across production payment batches.

Programmer Analyst Trainee Feb 2021 - May 2021
Cognizant *Coimbatore, India*

- Constructed a Spring Boot (Java) + Angular full-stack application to display bus routes and live location status for public transport users, supporting 5–10 active routes in test and demo environments.
- Crafted a relational database with 10+ route fields and REST APIs to handle 100–300 daily location updates, implementing GPS validation and normalization to reduce inconsistent records by 15–20%.

Supply Chain Data Analyst, Devi Scientific Enterprises Mar 2020 - Jan 2021
Devi Scientific Enterprises *Visakhapatnam, India*

- Analyzed 25K+ procurement orders and supplier sales by querying SQL databases to extract, join, and aggregate metrics, and used Pivot Tables & Power Query to summarize and identify cost-saving opportunities, improving sourcing efficiency by 12%.
- Optimized procurement workflows by tracking order lead times, supplier fulfillment rates, and stock availability using combined SQL queries and Excel reports, reducing supply chain lead times from 20 to 17 days.

Machine Learning Intern May 2019 – Jun 2019
ECIL *Hyderabad, India*

- Designed and implemented a real-time eye-tracking system on Raspberry Pi for wheelchair control using Python and OpenCV, integrating hardware sensors with computer vision pipelines.
- Employed frame skipping and optimized image processing to reduce detection and interpretation latency from 1.5s to 0.8s, and conducted user trials with wheelchair navigation demonstrating a 47% improvement in navigation speed and user control reliability.

Projects

FHIR-Aligned Healthcare Claims Analytics Pipeline – Portfolio Project Feb 2026

- Orchestrated automated ETL workflows using Azure Data Factory to build an end-to-end Azure pipeline transforming nested FHIR R4 JSON into a strict OMOP CDM v5.4 canonical schema, modeling 500K+ clinical and financial records.
- Enforced referential integrity across core OMOP domains by implementing deterministic UUID hashing and prefix normalization, preventing silent join failures across 211K+ claim lines in Azure SQL.
- Architected 5 OMOP Silver tables and 4 optimized Gold semantic views, eliminating Cartesian product row explosions by refactoring many-to-many LEFT JOIN logic with CTE-based pre-aggregation at the person_id grain.
- Delivered a production-ready BI semantic layer powering a real-time executive dashboard, reducing complex query execution time from 3–4 minutes to ≤ 3 second and stabilizing 30-day readmission and high-cost risk cohort analysis.

Award Record Extraction Scraper – UTA Capstone Project Jan 2024 – May 2024

- Built a Python-based data extraction pipeline using Retrieval-Augmented Generation (RAG) to scrape, parse, and structure award and fellowship records from websites, generating structured JSON outputs from unstructured descriptions for automated analysis.
- Optimized parsing and data retrieval logic to improve extraction accuracy by 20%, while implementing robust HTML handling, structured error logging, and failed-scrape recovery mechanisms to ensure reliable and consistent pipeline execution.

Job Recommender System – LinkedIn style & Analytics Sep 2023 – Dec 2023

- Built batch ETL pipelines in Apache Spark using DataFrame APIs, window functions, and Parquet to transform job posting and user interaction data into analytics-ready datasets.
- Executed a batch recommendation workflow in Spark combining feature engineering and Spark ML components to generate Top-N job recommendations for users with limited interaction history in standalone mode.

Banking System Database Design & Analytics Sep 2023 – Dec 2023

- Designed and normalized a relational database schema (8+ tables) modeling Customers, Accounts, Loans, and Transactions, ensuring 3NF compliance, referential integrity, and data consistency.
- Populated the database with 2,000+ simulated transaction records and developed SQL queries using joins, aggregations, and subqueries to generate reports on account balances, loan statuses, and branch-level summaries for simulated managerial analysis.

Technologies

Languages: Python (Advanced), SQL (Advanced), Java
Data Engineering: ETL/ELT Pipelines, Data Modeling (Star/Snowflake Schema), Apache Spark / PySpark, Pandas, NumPy, RAG
Cloud: Azure Pipelines, AWS (EC2, S3, RDS, Redshift)
Databases: MySQL, SQLite, ChromaDB
Tools: Power BI, Git/GitHub, Docker, CI/CD, Jenkins, Jira, REST APIs, PCF

Certification:

Microsoft Certified: Power BI Data Analyst Associate

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

Publications

- IJMTE 2020 [paper-link]: B. Somasekhar, Ch. Sai Kishore, **A. Pratyusha**, Nithinkumar. Sub-optimal Energy-Efficient Resource-Allocation to Maximize Energy Efficiency in OFDM System. In the International Journal of Management, Technology And Engineering, Volume X, Issue III, MARCH 2020, ISSN NO : 2249-7455, pg. 92.
- TEST Eng. & Mgmt. 2020 [paper-link]: V. Rajayalakshmi, B. Chandra Mouli, Ch. Sai Kishore, **A. Pratyusha**, P. Sai Surya Srinivas. Over Speed and Alcohol Detection SMS Alert System. In the TEST Engineering and Management, Vol. 83 Pg: 2329-2335, Issue: May-June 2020.

Education

Masters in Data Science | *University of Texas at Arlington Arlington, USA* Aug 2022 – May 2024

- Relevant Coursework: Machine Learning, Data Science, Data Analytics, Database Systems, Data Mining, Neural Networks, Probability and Statistics.

BTech in ECE | *Andhra University Visakhapatnam, India* Jun 2016 – Sep 2020